

# Tom Stesco

✉ tom.stesco@protonmail.com | ☎ +1 (416)-999-6759

## EDUCATION

### ETH ZURICH

#### MSC INTEGRATED BUILDING SYSTEMS

October 2018 | Zurich, Switzerland  
5.75 / 6.0 thesis grade

### UNIVERSITY OF WATERLOO

#### BASc HONOURS CIVIL ENGINEERING

June 2016 | Waterloo, ON  
With Distinction, 90% final term grade

## LINKS

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## PROGRAMMING

### EXPERT

Python

### PROFICIENT

C++ • SQL • Bash

### PRIOR EXPERIENCE

Java • Javascript

### TOOLS

Numpy • Pandas • Statsmodels

SKlearn • PyMC3 • CasADi

Plotly • Matplotlib • Jupyter lab

MySQL • BigQuery • BigTable

Docker • Kubernetes • GCP

GNU/Linux • Git

## COURSEWORK

### GRADUATE

Model Predictive Control

Mathematical Optimization

Computational Physics

Building Control and Automation

Computational Fluid Dynamics

Renewable Energy Technologies

Technology and Innovation Management

Innovation Leadership

Social Networks Research

### UNDERGRADUATE

Software Engineering Design Project

Building Science

Air Pollution Control

Bioprocess Engineering

Advanced Calculus

Differential Equations

Probability and Statistics

Linear Algebra

## EXPERIENCE

### ECOBEE | SENIOR DATA SCIENTIST

Jun 2021 – present | Toronto, ON

- Lead development of Model Predictive Control (MPC) optimization algorithms in C++ on embedded IoT devices to increase customer HVAC energy savings and improve thermal comfort.
- Solved IoT device temperature calibration regression model generalization issues by preprocessing training data and adding k-fold cross validation to feature engineering.
- Accelerated thermodynamics predictive modelling team cycle time from weeks to days by automating training, validation, and deployment in GCP using Python and Docker.
- Lead external research collaborations with 4+ groups related to HVAC controls and simulation software and 2 industry partnership MM\$ projects.

### ECOBEE | DATA SCIENTIST

Oct 2018 – Jun 2021 | Toronto, ON

- Scaled internal IoT experimentation platform from 0 to 150k+ devices running 10+ experiments simultaneously by automating build, deployment, and monitoring with C++, Bash, and Python.
- Lead development of [github.com/ecobee/building-controls-simulator](https://github.com/ecobee/building-controls-simulator), an open source Python library for HVAC control algorithm performance testing and presented at the **eSim2020** conference.
- Analyzed TB scale sensor and product data to make data driven recommendations and identify issues.

### ENERGY PROFILES LIMITED | ENERGY DATA ANALYST (INTERN)

Sep 2015 – Dec 2015 | Toronto, ON

- Developed equipment issue and input data error detection SQL queries using energy bill and submeter data.
- Added features and bug fixes to energy bill data ingestion pipeline in Java.

## RESEARCH

### AUTOMATIC CONTROL LAB | MASTER STUDENT

Sep 2017 – Oct 2018 | Zurich, Switzerland

- Worked with Dr. Annika Eichler on project combining Model Predictive Control (MPC) and online Bayesian learning for building controls.
- Developed Python library to model building thermodynamics and run MPC simulation experiments on HPC cluster.
- Developed online Bayesian Linear Discriminant Analysis (LDA) classifier for thermal comfort using PyMC3 with an MCMC algorithm.
- Developed predictive model of occupancy probability in Python using Markov chains and change point detection algorithm.

## AWARDS & PROGRAMS

2016 Excellence Scholarship & Opportunity Program

2016 Engineer of the Future Fund

2012 Merit Scholarship

2010 SHAD summer program

ETH Zurich

University of Waterloo

University of Waterloo

SHAD Canada